The Spectral Theorem

Recall: $T \in L(V)$ has a diagonal matrix rep. w.r.t. a basis iff the basis consists of eigenvectors of T.

The spectral theorem characterizes operators $T \in L(V)$ that are diagonalizable w.r.t. some ~~basis~~ orthonormal basis in V. These operators are the normal operators when $F = \mathbb{C}$ and self-adjoint operators when $F = \mathbb{R}$.

Thm Complex Spectral theorem

Suppose $F = \mathbb{C}$, $T \in L(V)$. Then, the following are equivalent:

(a) T is normal, $TT^* = T^*T$.

(b) V has an orthonormal basis consisting of eigenvectors of T.

(c) T has a diagonal matrix w.r.t. some orthonormal basis of V.

Proof: (b) $\equiv$ (c) follows from 5.41 (we've done it before).

Let (c) holds w.r.t. O.N.B. $\mathcal{B}_V$, i.e., $[T]_{\mathcal{B}_V}$ is diagonal.

$[T^*]_{\mathcal{B}_V}$ is also diagonal as $[T^*]_{\mathcal{B}_V} = \overline{[T]_{\mathcal{B}_V}}^T$.

Two diagonal matrices commute $\Rightarrow$ T & T^* commute $\Rightarrow TT^* = T^*T$, i.e., T is normal; (a) holds.

Now, let (a) holds, i.e. T be normal. By Schur's thm., $\exists$ an ONB $\mathcal{B}_V'$ w.r.t. T has an upper Δular matrix, i.e., $[T]_{\mathcal{B}_V'}$ is upper Δ.

Let $\mathcal{B}_V' = \{\bar{e}_1, \dots, \bar{e}_n\}$.

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Let $\left([T]_{\mathcal{B}_V'}\right)_{i,j} = \begin{cases} a_{ij} & , i \le j \\ 0 & , i > j \end{cases}$; $i,j \in \{1,\dots,n\}$.

We now prove that $[T]_{\mathcal{B}_V'}$ is diagonal.

Note that $\|T\bar{e}_1\|^2 = |a_{11}|^2$ &

$\|T^{*}\bar{e}_1\|^2 = \sum_j |a_{1j}|^2$.

$\because$ T is normal, $\|T^{*}\bar{e}_1\|^2 = \|T\bar{e}_1\|^2$.

$\therefore$ $a_{1j} = 0 \ \forall j > 1$.

Again, $\|T\bar{e}_2\|^2 = |a_{22}|^2$, $\|T^{*}\bar{e}_2\|^2 = \sum_j |a_{2j}|^2$.

$\because$ T is normal, $\|T^{*}\bar{e}_2\|^2 = \|T\bar{e}_2\|^2$,

$\therefore$ $a_{2j} = 0 \ \forall j > 2$.

(In general, continuing in this order gives us:
$\|T\bar{e}_i\|^2 = |a_{ii}|^2 = \|T^{*}\bar{e}_i\|^2$.)

$\vdots$

We arrive at conclusion that $[T]_{\mathcal{B}_V'}$ is diagonal;
ⓒ holds.

Thm. Let $V \neq \{0\}$ & $T \in L(V)$ is self-adjoint. Then T has an eigenvalue.

Thm. If $T \in L(V)$ is self-adjoint then $\exists$ an ONB in V w.r.t. T has diagonal matrix.
(The converse is also true when $F = \mathbb{R}$.)

Positive Ops. & Isometries

An operator $T \in \mathcal{L}(V)$ is called +ve, $T \geq 0$, if $T = T^*$ and $\langle T\bar{v}, \bar{v}\rangle \geq 0 \ \forall \ \bar{v} \in V$.

(Note: When $\mathbb{F} = \mathbb{C}$, $\langle T\bar{v}, \bar{v}\rangle \geq 0 \ \forall \ \bar{v} \in V \Rightarrow T = T^*$; as $\langle T\bar{v}, \bar{v}\rangle \in \mathbb{R} \ \forall \ \bar{v} \in V$ iff $T = T^*$)

Examples: If U is a subspace of V, then the orthogonal projection P_U is a +ve operator. The orthogonal projection of V onto U is $P_U \in \mathcal{L}(V)$ defined as: For $\bar{v} \in V$, write $\bar{v} = \bar{u} + \bar{w}$, where $\bar{u} \in U$ & $\bar{w} \in U^\perp$. Then $P_U \bar{v} = \bar{u}$.

$T \in \mathcal{L}(V)$ is self-adjoint & $b, c \in \mathbb{R}$ s.t. $b^2 < 4c$, then $T^2 + bT + c\mathbb{1} \geq 0$.

An operator R is called a square root of $T \in \mathcal{L}(V)$ if $R^2 = T$.

Thm. Characterization of +ve operators.
Let $T \in \mathcal{L}(V)$. Following are equivalent:

(a) $T \geq 0$ (T is +ve).
(b) $T = T^*$ (T is self-adjoint) and all the eigenvalues of T are non-negative.
(c) T has a +ve square root.
(d) T has a self-adjoint square root.
(e) $\exists \ R \in \mathcal{L}(V)$ s.t. $T = R^* R$.

Proof: Let (a) holds. (a) $\Rightarrow$ T is self-adjoint by def^n.
(a) $\Rightarrow$ $\langle T\bar{v}_\lambda, \bar{v}_\lambda\rangle = \langle \bar{v}_\lambda, T\bar{v}_\lambda\rangle = \lambda\langle \bar{v}_\lambda, \bar{v}_\lambda\rangle \geq 0$, where $\bar{v}_\lambda$ is ~~eigenvalue of~~ eigenvector corresponding to λ.
i.e., $\lambda \geq 0$. So, (a) $\Rightarrow$ (b).

By spectral thm, T is diagonal w.r.t. ONB of eigenvectors & diagonal entries are eigenvalues.
Let $\mathcal{B}_V = \{\bar{e}_1, \ldots, \bar{e}_n\}$ be ONB ~~of~~ consisting eigenvectors $\bar{e}_i$, $i \in \{1, \ldots, n\}$, corresponding to eigenvalues λ_i, $i \in \{1, \ldots, n\}$, of T.

$T\bar{e}_i = \lambda_i \bar{e}_i \quad \forall\, i \in \{1, \ldots, n\}$.

$$[T]_{\mathcal{B}_V} = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

Let $R \in \mathcal{L}(V)$ be s.t. $R\bar{e}_i = \sqrt{\lambda_i}\,\bar{e}_i$, $R \geq 0$.
where $\sqrt{\lambda_i} \geq 0$ as $\lambda_i \geq 0$.
Also, $T = R^2$ as $R^2\bar{e}_i = \lambda_i e_i = Te_i \quad \forall\, i \in \{1, \ldots, n\}$;
(c) holds. (c) $\Rightarrow$ (d) by def^n. (d) $\Rightarrow$ (e) as $R^* = R$.
(e) $\Rightarrow$ (a) as $T = R^*R = (R^*R)^* = T^*$ and
$\langle T\bar{v}, \bar{v}\rangle = \langle R^*R\bar{v}, \bar{v}\rangle = \langle R\bar{v}, R\bar{v}\rangle = \|R\bar{v}\| \geq 0$.

Thm. Every +ve operator $T \in \mathcal{L}(V)$ has a unique +ve square root.

Proof:

Isometries

- An operator $S \in L(V)$ is called an isometry if $\|S\bar{v}\| = \|\bar{v}\| \ \forall \ \bar{v} \in V$.

→ An operator is an isometry if it preserves norms.

Examples:

(a) $\lambda \mathbb{1}$ s.t. $|\lambda| = 1$, for $\lambda \in F$, is an isometry.

(b) Let $\lambda_j \in F$, $j \in \{1, \dots, n\}$, s.t. $|\lambda_i| = 1$. $S \in L(V)$ satisfies $S\bar{e}_j = \lambda_j \bar{e}_j \ \forall \ j \in \{1, \dots, n\}$, for some ONB $\mathcal{B}_V = \{\bar{e}_1, \dots, e_n\}$ of V. Prove S is isometry.

Proof: Say $\bar{v} \in V$. $\bar{v} = \sum_{j=1}^{n} \langle \bar{v}, \bar{e}_j \rangle \bar{e}_j$,

$$\|\bar{v}\|^2 = \sum_j |\langle \bar{v}, \bar{e}_j \rangle|^2$$

$$S\bar{v} = \sum_j \langle \bar{v}, \bar{e}_j \rangle S\bar{e}_j = \sum_j \lambda_j \langle \bar{v}, \bar{e}_j \rangle \bar{e}_j.$$

$$\|S\bar{v}\|^2 = \sum_j |\lambda_j|^2 |\langle \bar{v}, \bar{e}_j \rangle|^2 = \sum_j |\langle \bar{v}, \bar{e}_j \rangle|^2 = \|\bar{v}\|^2.$$

Thm. Let $S \in L(V)$. The following are equivalent:

(a) S is an isometry.

(b) $\langle S\bar{u}, S\bar{v} \rangle = \langle \bar{u}, \bar{v} \rangle \ \forall \ \bar{u}, \bar{v} \in V$.

(c) $\{S\bar{e}_1, \dots, S\bar{e}_n\}$ is orthonormal for every orthonormal set of vectors $\{\bar{e}_1, \dots, \bar{e}_n\} \subset V$.

(d) $\exists$ an orthonormal basis $\{\bar{e}_1, \dots, \bar{e}_n\}$ of V s.t. $S\bar{e}_1, \dots, S\bar{e}_n$ is orthonormal.

(e) ~~$SS^* = \mathbb{1}$~~ $S^*S = \mathbb{1}$

(f) $SS^* = 1$

(g) S^* is an isometry.

(h) S is invertible, and $S^{-1} = S^*$.

Proof:

Thm. Suppose V is a complex inner product space & $S \in \mathcal{L}(V)$. Then the following are equiv:

(a) S is an isometry.

(b) $\exists$ an ONB of V consisting of eigenvectors of S whose corresponding eigenvalues all have absolute value 1.

Proof: $(\lambda_j, \bar{e}_j)$, $S\bar{e}_j = \lambda_j \bar{e}_j$, $S^*S\bar{e}_j = \bar{\lambda}_j \lambda_j e_j = |\lambda_j| \bar{e}_j$

$S^*S = 1$, $S^*S\bar{e}_j = \bar{e}_j$

$|\lambda_j| = 1$.